



ACADEMIC CITY
UNIVERSITY

BSc

COMPUTER ENGINEERING

Computer Engineering at ACity

In ACity, our Computer Engineering program is designed to provide those passionate about computing, computer hardware and peripheral design as well as software engineering with the requisite tools and skills needed for further advancement in digital technology, computer networking, and computer systems. The depth of our program ensures that students at graduation are at par with contemporaries from similarly placed institutions.

Finally, this curriculum delineates pathways for students hoping to pursue advanced studies in Computer Engineering by ensuring that specific electives allow them to focus on specific areas of concentration.



Grooming Tomorrow's Computer Gurus

A simple abacus may have inspired its invention, but after gaining its greatness in the last two decades of the 20th century, computers have taken the lead in technological developments across the globe in recent years.

With the ever-evolving area of complex computer systems, hardware and software, the need to groom young talents for technological advancement have become necessary for the rapid growth of industry.

These young creative minds, known professionally as Computer Engineers, will occupy the key role as software developers and hardware engineers among other roles in many industries.

Top Careers In Computer Engineering

The US Bureau of Labor Statistics projects a 5% growth for computer engineering jobs through 2026. This statistic represents significant opportunities for computer engineering students as they prepare for the world of work. What jobs await you in the future? Check out some top options

- ✦ Computer Hardware Engineer
- ✦ Computer Network Architect
- ✦ Computer Systems Analyst
- ✦ Cyber Security Analyst
- ✦ Software Developer
- ✦ Big Data Engineer

Why ACity?

Home to
students from over
23 countries
across Africa

A fully digitized
campus with the fastest
internet speed
in a Ghanaian university

70%
practical lessons

**Experienced
Faculty**
with Industry Expertise

Every **ACitizen** has
the opportunity to do a
work placement
every vacation

Student-To-Faculty Ratio
5:1

Houses a
Digital Makerspace
and a **Technology &
Entrepreneurial Center**
for **Creative freedom
& Innovation**

Our Unique Learning Pillars



Experiential Learning

Hands-on learning to prepare students to readily apply concepts, to easily integrate into the workspace.



Contextual Learning

Solving real grass-root problems to expose students to the local context and develop empathy towards the continent's progress.



Unified Learning

A project-based approach that combines concepts across courses to connect the dots and enable unified learning.



Extensional Learning

Arms students with a viable toolkit to help them confront real-life issues, they may not have encountered during their academic life, squarely.

Entry Requirements

WASSCE/NECO

Minimum C6 in 6 subjects including 3 core subjects (English, Maths, Integrated/General Science) and 3 elective subjects. (Physics, Elective Maths + Chemistry OR any other elective relevant to the chosen Program)

SSCE

Minimum D or a pass in 6 subjects including 3 core subjects (English, Maths, Integrated/General Science) and 3 elective subjects. (Physics, Elective Maths + Chemistry OR any other elective relevant to the chosen Program)

IGCSE O-LEVEL & A-LEVEL

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and 3 passes in the A-Levels. (Elective/Add/Further Maths and Physics mandatory)

ENGLISH IB

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and a minimum score of 4 points in 3 Higher Level (HL) subjects. (Elective/Add/Further Maths and Physics mandatory)

FRENCH IB

Minimum of 50% overall average pass. (subject to approval NAB) (Maths, English and Physics mandatory)

AMERICAN HIGH SCHOOL

Minimum GPA of 3.0 (Maths, English and Physics mandatory)

COURSE OUTLINE

SEMESTER 1

Course Name

Communication Skills
French Language
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar I
Introduction to Engineering
Introduction to Programming with Python
Physical Sciences
Pre-Calculus (with MATLAB)
Technology and Society

SEMESTER 2

Course Name

Analytic Geometry and Calculus I (with MATLAB)
Basic Electronics
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar II
Introduction to Multidisciplinary Design
Logic and Critical Thinking
Programming in C
Sensors, Measurements and Instrumentation
Text and Meaning

SEMESTER 3

Course Name

Analytic Geometry and Calculus II (with MATLAB)
Circuit Theory
Computer Systems Engineering
Data Structures and Algorithms
Fundamentals of Innovation and Entrepreneurship (FIE) I
Fundamentals of Logic Design
Introduction to Material Science and Engineering
Leadership Seminar I

SEMESTER 4

Course Name

African Studies
Applied Linear Algebra (with MATLAB)
Computer Architecture and Organisation
Digital Systems Design
Fundamentals of Innovation and Entrepreneurship (FIE) II
Microelectronics
Probability, Statistics and Reliability (with MATLAB)

SEMESTER 5

Course Name

Computer Networks
Differential Equations (with MATLAB)
Discrete Mathematics (with MATLAB)
Leadership Seminar II

SEMESTER 5

Course Name

Microprocessors and Microcontrollers
Signals and Systems
Operating Systems
System Dynamics

SEMESTER 6

Course Name

Automatic Control Systems
Digital Signal Processing
Embedded Systems
Industry Internship

SEMESTER 6

Course Name

Mechatronics
Software Engineering
Project Management, Engineering
Economics and Risk Analysis

SEMESTER 7

Course Name

Computer Security
Introduction to Artificial
Intelligence
Project Phase I
Technical Elective - I
Technical Elective - II

SEMESTER 8

Course Name

Environmental Science and
Engineering
Professional Ethics and Values
Project Phase II
Technical Elective - III
Technical Elective - IV

ELECTIVES

Biomedical Engineering

Biomedical Instrumentation
Biomedical Signal Analysis
Real-Time Digital Signal Processing
Wireless Sensor Networks

Telecommunication

Advanced Digital Signal Processing
High Speed Networks
Remote Sensing
Telecommunication Systems
Modelling and Simulation

Intelligent Systems: Controls and Automation

Advanced Control Systems
Digital Image Processing
Machine Learning
Modeling and Analysis of
Mechatronic Systems

Intelligent Systems: Robotics and Mechatronics

Advanced Control Systems
Industrial Automation and Robotics
Machine Learning
Modeling and Analysis of Mechatronic
Systems

Renewable Energy Systems

Smart Grid
Solid State Electronic Devices
Energy Management for a
Sustainable Future
Green Electronics